

Looking towards the Future of BIM in South Korea Towards AI-Enhanced BIM

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【Abstract】 This paper provides an overview of South Korea's 20-year journey in adopting building information modeling (BIM) and future direction. It first discusses the six phases of BIM adoption in South Korea, starting from the use of BIM as a marketing tool to its current intelligent BIM phase. The government's support for BIM-related research and development projects is also highlighted, with a focus on the artificial intelligence (AI)-based architectural design automation project. As the future direction, it explores the integration of AI with BIM in both local and global contexts. The paper presents AI-powered architectural design methods, including AI-powered early architectural design generation and architectural detailing. Compared to AI-based early architectural design generation, architectural detailing is an unexplored research topic. This paper introduces two AI- and BIM-based architectural detailing methods, being developed at Yonsei University: namely, BIM library transplant and Natural language-based Architectural Detailing through Interaction with AI (NADIA). These methods demonstrate how AI-enhanced BIM can enable architects to interactively develop building details using a language model as a conversational AI and a knowledge base, and a BIM authoring tool as a design platform, in the near future.

【Keywords】 building information modeling (BIM); artificial intelligence (AI); South Korea, BIM adoption; BIM utilization level (BUL); Natural language-based Architectural Detailing through Interaction with AI (NADIA); BIM library transplant

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Introduction

South Korea, like many other countries, has a long history of relying on traditional two-dimensional (2D) drawings, which has led to some resistance to change. While not all construction projects deploy building information modeling (BIM) yet, it has become the norm for most major projects in South Korea today. This marks a remarkable leap considering the limited awareness of BIM just two decades ago. Nowadays, BIM is a familiar term to almost everyone, regardless of their personal usage.

This paper presents a brief overview of South Korea's 20-year journey in adopting BIM, which unfolds in six phases. Furthermore, it highlights recent

advancements in integrating artificial intelligence (AI) and BIM, both within South Korea and on a global scale. Finally, the paper introduces two cutting-edge AI-powered BIM detailing methods developed at Yonsei University: the BIM library transplant and an interactive architectural detailing framework using natural language.

1 Background

The building informatics group (BIG) at Yonsei University has been conducting a biennial survey on BIM adoption since 2008. The survey results reveal that Korea started using BIM as early as the late 1900s. In a post-survey interview, it was discovered that the initial adopters

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of BIM in Korea were small architectural firms. These firms faced financial constraints during the late 1990s economic crisis and, as a result, turned to BIM as a means to streamline building design with minimal staff. However, it was not until 2008 that the number of BIM users experienced a significant surge.

Even prior to 2008, there were small-scale conferences and gatherings focused on BIM. One notable example is a BIM seminar series hosted by the international alliance for interoperability (IAI) Korea. IAI is now known as buildingSMART[1]. In 2008, South Korea hosted its first large-scale BIM gathering, the buildingSMART Forum 2008, which attracted over 1,000 attendees. The industry's interest was further fueled by the government's drive for BIM. The Ministry of Land, Infrastructure, and Transport of Korea (MOLIT), along with the Public Procurement Service (PPS), announced their goal to implement the BIM roadmap, commencing with two to three pilot projects in 2010 and ultimately mandating BIM for all public building projects over 50 billion Korean won (KRW) by 2016. Chapter 8 of the BIM Handbook, 3rd edition, provides a detailed discussion of BIM mandates around the world, including Korea^[2].

However, the journey was challenging. The number of BIM projects experienced significant fluctuations due to economic conditions. Additionally, despite the industry's understanding of the benefits of BIM adoption, there were certain resistances encountered. Barriers to BIM adoption included the cost of investment, learning curve, lack of executive buy-in, shifting of liability among project participants, poor collaboration among stakeholders, limited interoperability among BIM software, reluctance to openly share information, and other factors^[3]. Nevertheless, despite these ups and downs and the presence of barriers, BIM adoption in Korea has steadily progressed across various sectors within the construction industry, encompassing both large and small firms. It is important to note that BIM adoption in Korea remains an ongoing process. On June 12, 2023, MOLIT announced an updated goal to mandate BIM on all public projects, including both building and infrastructure projects with a budget exceeding 100 billion KRW, by 2030.

2 Status of BIM adoption in South Korea

2.1 Six waves of BIM adoption

Many people initially thought that the transition from 2D-drawing-based practice to BIM would be a one-giant-leap process^[4]. However, many years of BIM adoption studies show that the transition from 2D computer-aided design and drafting (CADD) to BIM involves transformation in a series of incremental steps rather than a single jump. Our BIM adoption study for the past 20 years reveals that BIM implementation has taken place and is still ongoing, primarily unfolding in six distinct phases. A BIM maturity model developed based on this six phases is the BIM utilization level (BUL) model^[5]. The following paragraphs describe these six phases:

Visual BIM: In the mid-2000s, in Korea and also in many other countries, BIM was primarily utilized as a marketing and visualization tool. Since the information within the BIM models was seldom utilized during this period, people commonly referred to this utilization of BIM as “BIM wash”^[6].

Conversion BIM: The next step was to utilize BIM for checking the consistency of drawings and identifying design errors through a “CADD to BIM conversion” process, which involved converting 2D drawings into BIM models. This conversion process typically occurs during the final design phase or the preconstruction phase. One of the critical drawbacks of this approach is that the generated design BIM models and related information are generally not actively utilized or consistently maintained throughout the construction phase, which limits its overall benefits. This approach was prevalent during the late 2000s and continues to be commonly practiced in numerous countries, including Korea.

Two-track BIM: In the early 2010s, many projects began utilizing BIM in a slightly more advanced manner than conversion BIM. BIM was employed not only for design review through 2D-to-BIM conversion during the design and preconstruction phases but also throughout the construction phase. However, drawings continued to serve as the primary communication medium. BIM was used as a supplementary method for designing, reviewing, and constructing complex geometric or engineering-wide areas. As both BIM and drawings were utilized during this phase,

it became commonly known as “two-track BIM” or “parallel BIM.”

Integrated BIM: It was not until the late 2010s that a significant number of projects began to use BIM the way BIM pioneers envisioned it should be used. During this period, numerous projects in Korea deployed BIM in a fully integrated manner. By “fully integrated,” we mean that the majority of project participants utilizes BIM as the primary communication medium although some individuals still rely on traditional drawings. In this method, most drawings are produced within the BIM environment as a byproduct of BIM activities.

Lean BIM: With the growing interest in Construction 4.0 and smart construction in the 2020s, there was a push to transform the construction industry to resemble the manufacturing sector more closely. This transformation involves various initiatives, such as modular construction, the internet of things (IoT), 3D printing, off-site construction, the integration of BIM and lean methods for construction management, and the utilization of augmented reality (AR), virtual reality (VR), mixed reality (MR), and robotic construction technologies, such as robot dogs walking around the site. The “lean BIM^[7,8]” phase is also referred to as “connected BIM^[9]” since these advancements rely on the seamless connection and exchange of signals and information between BIM models and other systems, such as 3D printers or robotic dogs.

Intelligent BIM: The latest trend that we have observed is the integration of data analytics, artificial intelligence, and BIM. The primary objective of utilizing BIM is to enable informed decision-making. In order to achieve this, the integration of public data and BIM data becomes crucial as valuable sources of information. By harnessing big data and combining it with digital twins, BIM is expanded from individual projects to encompass large-scale megaprojects, including smart cities, airports, seaports, and even aerospace endeavors such as the Moon base camp project. This phase is commonly referred to as the intelligent BIM phase.

Using the BUL model, the progress of BIM adoption in South Korea has been monitored every two years since 2016. As depicted in the Figure 1, conversion BIM and two-track BIM remain the most prevalent methods of

BIM utilization. However, the figure also demonstrates a noticeable trend of a progressively rising percentage of advanced approaches to BIM implementation.

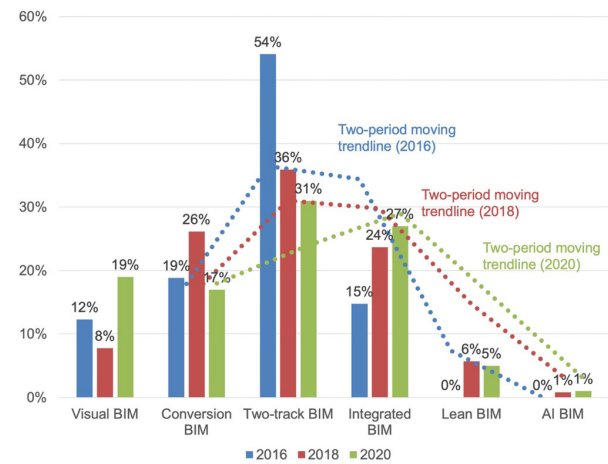


Figure 1 Distributions of BIM utilization levels in South Korea from 2016 to 2020

2.2 Government-funded BIM R&D projects

There was also strong support for developing BIM-related new technology research and development by the Korean government. The number and budget of BIM-related R&D projects have been continuously increasing. As of 2022, there are approximately 200 BIM-related R&D projects, with a total budget of 167 billion Korean won. Major national BIM R&D projects in South Korea, with a budget over 10 billion KRW from 2006 to date, encompass the following initiatives: the virtual construction project (2006-2011), the open BIM project phase 1 (2012-2016) and phase 2 (2017-2021), the open BIM solution development project (2011-2013), the rail BIM project phase 1 (2015-2018) and phase 2 (2020-2024), the smart construction project (for highways) (2020-2025), and the AI-based architectural design automation project (2021-2025).

Among these, this paper delves into the AI-based architectural design automation project focusing on AI-powered early architectural design generation and detailing as a future direction.

3 Future direction: AI-powered architectural design

3.1 AI-powered early architectural design generation

The integration of AI and BIM will be a natural next step in the future as we discussed related to the BUL

model. In 2016, when Go master Yi Sedol was defeated by AlphaGo, numerous surveys were conducted to determine which jobs would persist in the future and which ones would not. Many surveys conducted at that time suggested that AI would be unable to easily replace jobs that necessitate creativity, such as painters, sculptors, and writers.

However, with the emergence of generative adversarial networks (GANs), it has become possible for AI to generate paintings and designs. One of the earliest AI models that generates architectural plans is ArchiGAN, which was developed in 2019. SpaceMaker is an enhanced version of ArchiGAN that was published in 2020. Despite technological advancements, the outputs of ArchiGAN and SpaceMaker were bitmap images that could not be imported into a CAD system. In 2020, HouseGAN, a GAN-based AI that generates architectural plans based on a bubble diagram, was able to produce architectural plans in a vector format. HouseGAN was further refined into HouseGAN++ in 2021.

There have also been academic efforts in South Korea to develop AI-based tools for early architectural design. These include FacadeGAN, a GAN-based model for generating facade designs, developed at Yonsei University^[10], as well as AIBIM MassMaker, SpaceMaker, and Pre-Design Cost Estimator developed by Kyungpook National University^[11].

Nowadays, several AI models for early architectural designs have been commercialized. One such system is Finch3D^[12], which can automatically generate an architectural plan optimized for various input factors. Architectures.com offers a similar service^[13]. Another system called PlanFinder automatically generates an ideal floor plan that fits within a given area^[14].

In Korea, there are numerous companies specializing in AI-based automated architectural design. One example is TenEleven, which focuses on automated design layout considering factors such as building codes and regulations, and economic feasibility^[15]. HowBuild utilizes AI to generate schematic building designs for early cost estimates^[16]. SpaceWalk provides a service called Landbook, which is an AI-powered automated feasibility analysis service for real estate developers and

individuals interested in constructing a building^[17]. Flexity focuses on generating practical design solutions, while other companies may emphasize early cost estimates or feasibility analysis^[18].

The examples mentioned above highlight the use of AI in generating schematic designs. However, with recent advancements in large language-image AI models, AI is also being employed in the development of conceptual designs. An excellent illustration of this is the award-winning digital art piece “Space Opera Theater^[19]” created by Jason Allen using the image-generation AI, called Midjourney^[20]. While “Space Opera Theater” wasn’t specifically intended to showcase the potential use of AI in conceptual designs for building interiors or exteriors, it has inspired numerous other examples of utilizing AI for generating conceptual architectural designs^[21].

3.2 AI-powered architectural detailing

Although AI has been actively used for early design generation, the use of AI for developing architectural details still remains unexplored. This section briefly introduces some of the efforts at Yonsei University to develop AI-based architectural detailing methods. These methods are being developed as part of the R&D project to develop “Artificial Intelligence-Based Architectural Design Automation Technologies,” which is funded by the Ministry of Land, Infrastructure, and Transport of Korea from 2021^[11].

One method is called “BIM library transplant.” It takes a BIM model with a high level of detail as a donor model and replaces the BIM libraries in a BIM model with a low level of detail with those in the donor model. The essence of this method lies in the “library matching” module, which can use either rule-based or machine-learning approaches, or a mixture of both. Currently, the BIM library transplant has been explored using several methods: RhinoInsideRevit and the Revit application programming interface (API), potentially in conjunction with the utilization of a graph convolutional network (Figure 2a).

Another method is “Natural language-based Architectural Detailing through Interaction with AI (NADIA)^[22].” Early pioneers of AI envisioned that

architects would be able to generate BIM by interactively talking to a computer. NADIA enables architects to interactively develop details of a building using a language model as a conversational AI and a knowledge base, and a BIM authoring tool as a design platform. Currently, NADIA has been implemented using GPT version 3.5 and the Revit API (Figure 2b).

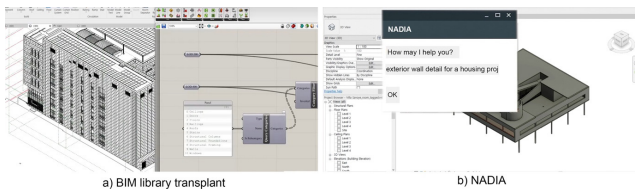


Figure 2 AI-enhanced BIM detailing: BIM library plant and NADIA

4 Conclusion

This paper describes how South Korea has made significant progress in adopting BIM over the past 20 years through six phases of the BIM model, despite initial resistance and challenges. These six phases of BIM adoption, ranging from visual BIM to intelligent BIM, demonstrate the gradual transformation and increasing utilization of BIM in various sectors of the construction industry. The government's mandate of BIM on public projects was a key driver for BIM adoption, and substantial investments in BIM-related R&D projects have further facilitated the integration of BIM and AI. This integration has led to advancements in AI-powered architectural design, enabling automated generation of early designs and conceptual designs. Several examples of such initiatives from academia and the industry have been explained in the paper.

However, the development of AI-based architectural detailing has received relatively little exploration. To illustrate AI-based architectural detailing, this paper introduces two examples: the BIM library transplant and NADIA, currently being developed by Yonsei University. The ongoing advancements and future directions of AI in architectural design and detailing show great promise for enhancing efficiency, decision-making, and collaboration in the construction industry by leveraging the synergies between BIM and AI.

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